

SPUTNIK 1

EXTERIOR

Aluminium sphere
23in (58cm) in
diameter

Aerial
transmitters
9ft (3m) long

SYMBOLIC SPHERE

The sole function of the satellite was to announce to the world a triumph of Soviet technology. Its signal, transmitted while in orbit, announced the start of the space age.

THE FIRST ARTIFICIAL EARTH SATELLITE, Sputnik 1, was launched from the Baikanur Cosmodrome near Tyuratam, Kazakhstan, late on the evening of October 4, 1957. To minimize the chances of failure, the Soviet team had chosen to keep the satellite simple, far lighter than the maximum load that the Soviet R-7 rocket was capable of lifting into space. Watching from a bunker as the huge rocket launched, Sergei Korolëv was almost incredulous at his own success. "Is this really all?" he said. "Have we really done it?" As Sputnik 1 circled some 500 miles (800km) above the Earth, emitting beeps from its transmitters, the Soviet people were euphoric. As for the American reaction, future president Lyndon B. Johnson recalled "the profound shock of realizing that it might be possible for another nation to achieve technological superiority over this great country of ours."

EXPLODED VIEW

INSIDE SPUTNIK

Sputnik 1 was an aluminium sphere containing two radio transmitters, a series of silver-zinc batteries, and a thermometer. It weighed 184lb (84kg). Traveling at 18,000mph (30,000kph), it orbited the Earth every 96 minutes.

the marshal then fell victim to Stalin in the great purge that decimated the Red Army officer corps. Korolëv was among many scientists sent to the Gulag on absurd charges of subversion. Soviet rocket research went into temporary decline.

The end of World War II left the Soviets in control of the V-2 production site at Nordhausen. They also kidnapped a number of generally lower-level scientists and technicians who had worked on the German rocket program. Information from these sources was used by Korolëv, now released from the Gulag, and other Soviet rocket experts such as Valentin Glushko to advance the development of the Soviet rocket program from artillery expertise initially

begun in World War II. By the early 1950s the Soviets had rocket motors capable of generating over 200,000lb (90,000kg) of thrust – the Redstone's motor gave 75,000lb (35,000kg).

First satellite

Whereas the Americans concentrated much of their energy on developing a nuclear bomber force, from 1952 the Soviets put the lion's share of their resources into building a nuclear-armed intercontinental ballistic missile (ICBM). The goal, as defined by Soviet Air Force chief Marshal Zhigarev, was to make "long-range, reliable rockets capable of hitting the American continent."

In 1955–56 a new launch complex, known as the Baikonur Cosmo-drome, was built in the Soviet republic of Kazakhstan. Here the Soviets set out to test a rocket that would astonish the world. By grouping five rocket motors together, Korolëv gave their R-7 Semyorka almost 1 million lb (455,000kg) of thrust. It had a range of 4,000 miles (6,400km), making it the world's first ICBM, and could potentially lift a one-ton payload into space.

After many delays, test launches of the R-7 began in May 1957. The first eight attempts either resulted in an aborted launch or explosion some time after liftoff. Korolëv was close to nervous collapse by the time a fully successful flight was achieved,

on August 21, 1957. After one repeat performance, Soviet leader Nikita Khrushchëv gave the order for a satellite mission. The launch of Sputnik 1 on October 4 had a propaganda impact that could only be described as sensational. The Soviet leadership immediately gave instructions for a follow-up. The launch of a dog into orbit in Sputnik 2 came exactly one month later.

SPUTNIK 3

Launched on May 15, 1958, Sputnik 3 was a conically-shaped vehicle measuring 11ft 8in x 5ft 8in (3.6m x 1.7m), and weighing 2,919lb (1,327kg). It was powered by solar panels embedded around the base of the main body.



SPACE DOG

The ill-fated Laika, the first living creature to travel into Earth orbit, demonstrates its space pod. The dog paid for worldwide celebrity status with its life, dying when Sputnik 2 overheated.

